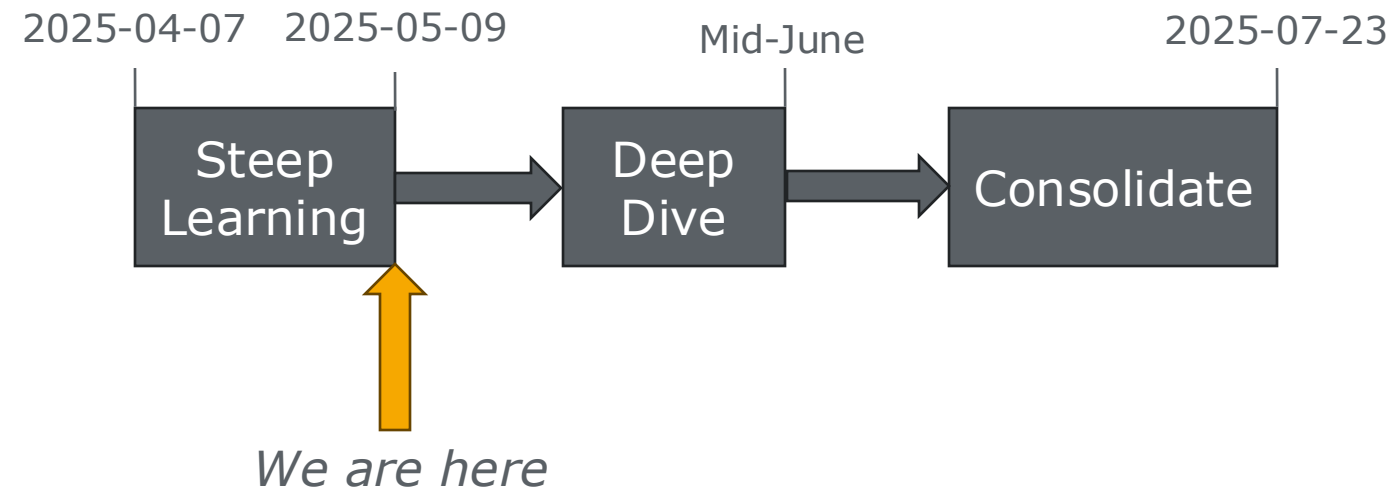




# OpenSmalltalk VM vs TruffleSqueak

Comparing Runtime Support for  
Toolbuilding Across System Boundaries

WORKING  
TITLE

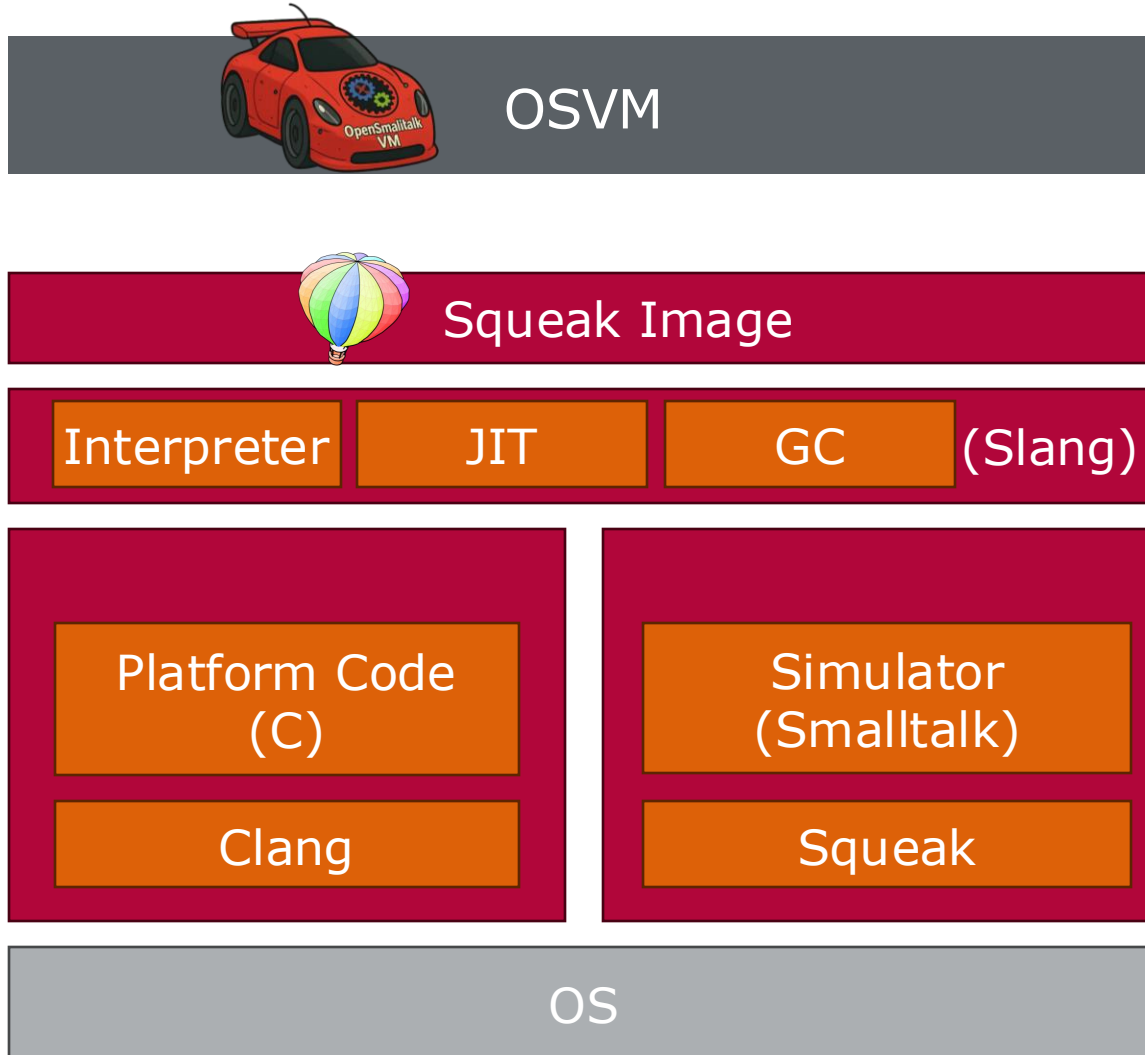


Master Project 2025

**Marius Dörbandt, Oliver Heß, Antony Kamp, Christoph Thiede**

Advisors: Marcel Taeumel, Jens Lincke, Robert Hirschfeld

# Background: Stacks



# Background: Stacks



OSVM

TSVM



Squeak Image

Interpreter

JIT

GC

(Slang)

TruffleSqueak (Java)

GraalPy

...

Platform Code  
(C)

Clang

Simulator  
(Smalltalk)

Squeak

Truffle (Java)

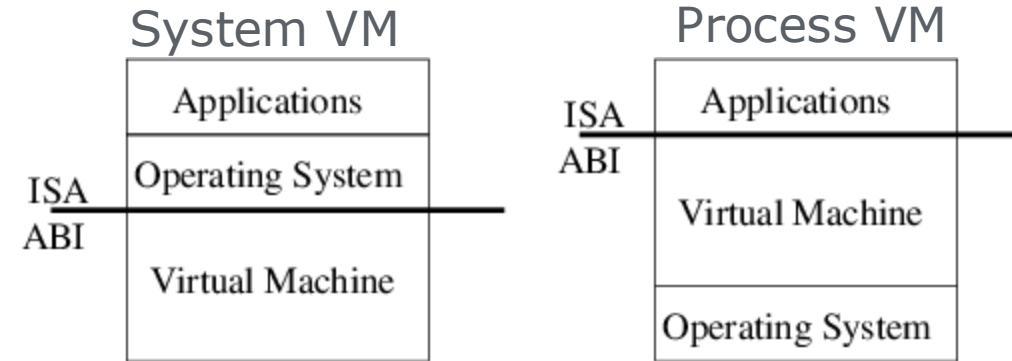
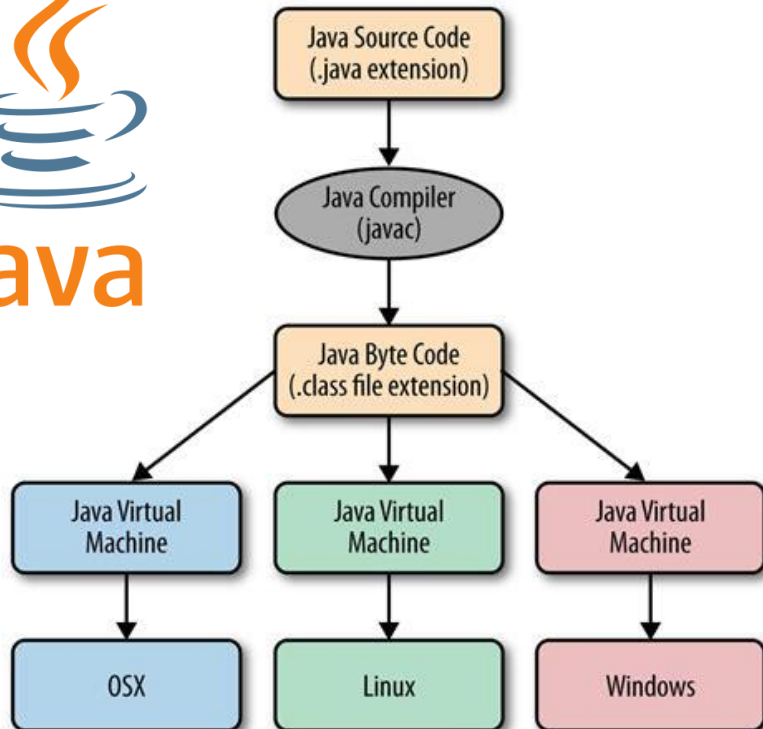
Graal Compiler

Hotspot Java VM

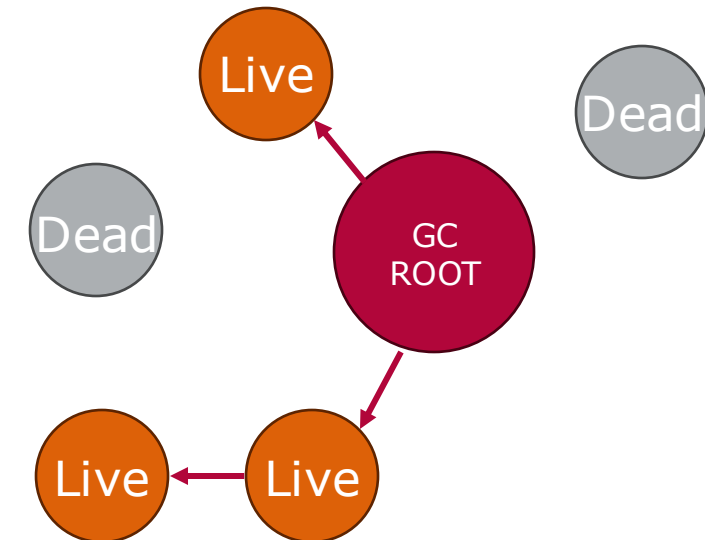
Native  
Image

OS

# Background: TSVM – General Concepts



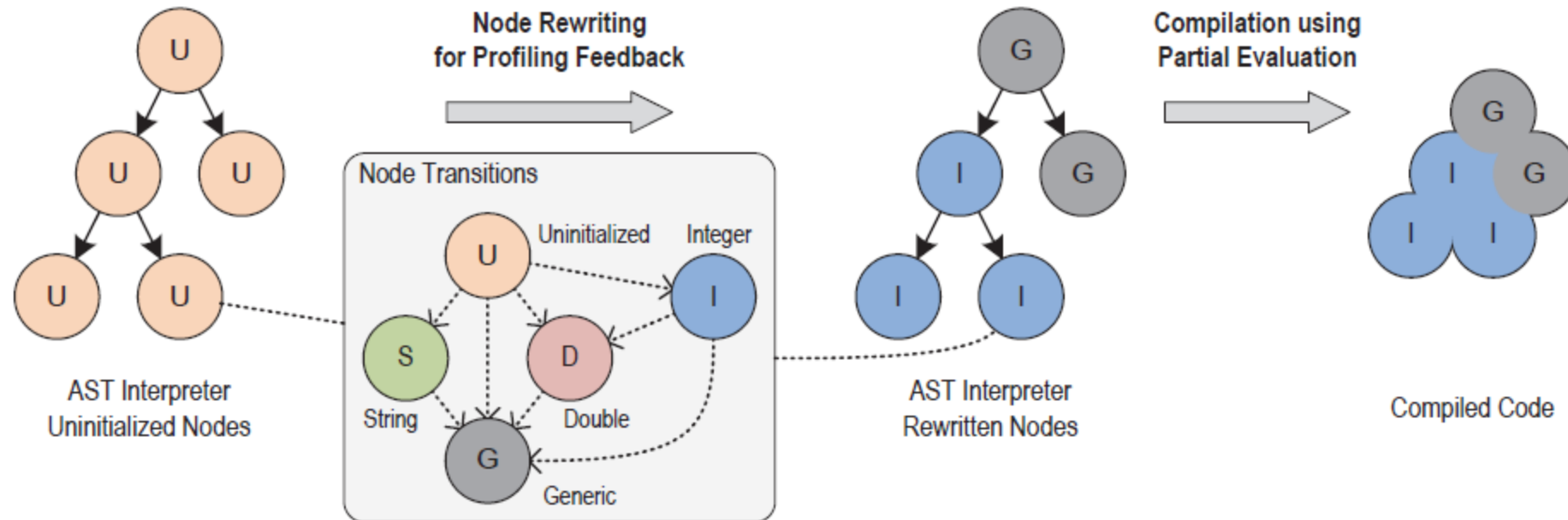
Andersen, Rasmus & Vinter, Brian. (2008). The Scientific Byte Code Virtual Machine.



# Background: TSVM – GraalVM & Truffle



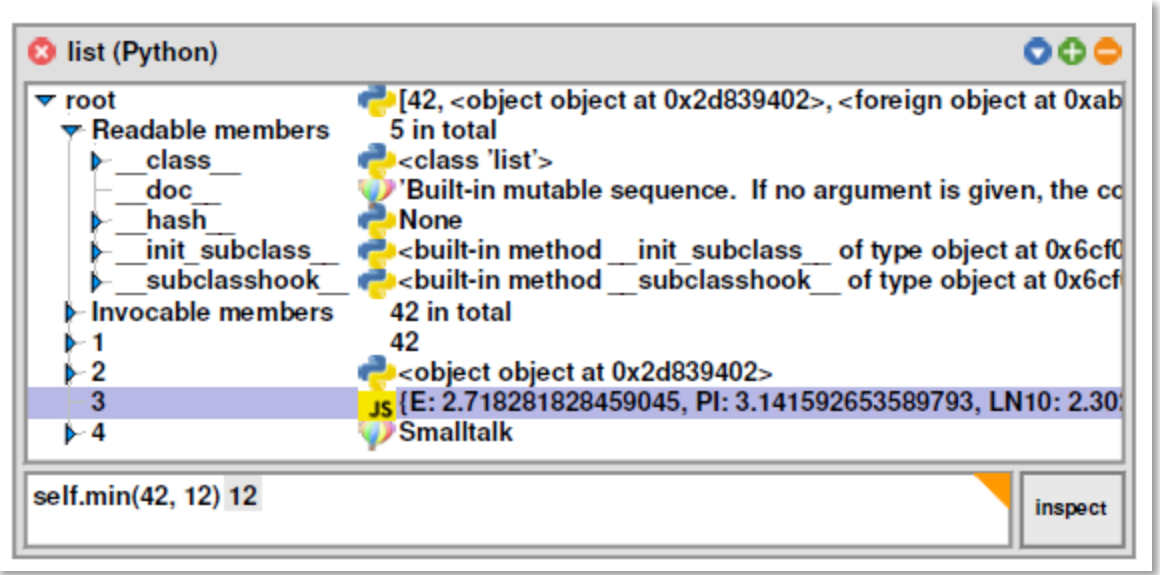
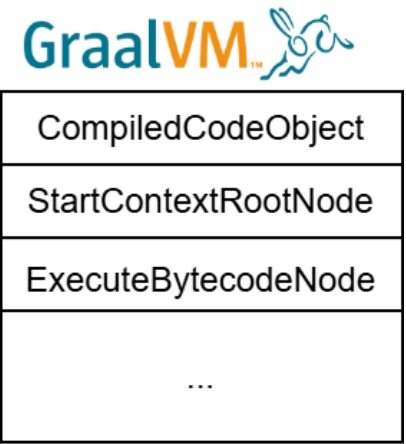
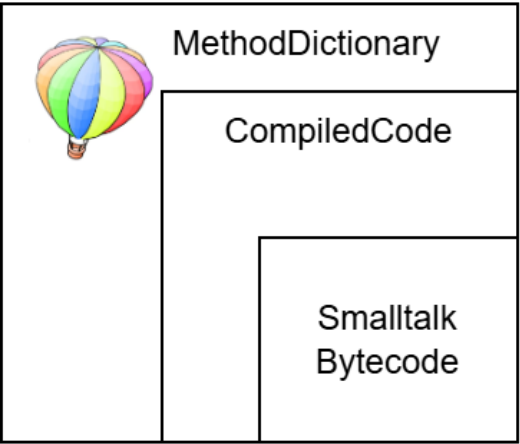
GraalVM™



**Figure 2.** Node rewriting for profiling feedback and partial evaluation lead to aggressively specialized machine code.

Würthinger, Thomas, et al. (2013). One VM to rule them all

# Background: TSVM – TruffleSqueak



Niephaus, Fabio. (2021). Exploratory Tool-Building Platforms for Polyglot Virtual Machines.

# Background: OSVM – JIT



```
Array>>sum
| sum |
sum := 0.
1 to: self size do: [:i |
    sum := sum + (self at: i)].
^ sum
```

```
41 <50> pushConstant: 0
42 <D0> popIntoTemp: 0
43 <51> pushConstant: 1
44 <D1> popIntoTemp: 1
45 <4C> self
46 <72> send: size
47 <D2> popIntoTemp: 2
48 <41> pushTemp: 1
49 <42> pushTemp: 2
50 <64> send: <=
51 <EF 0E> jumpFalse: 67
53 <40> pushTemp: 0
54 <4C> self
55 <41> pushTemp: 1
56 <90> send: at2:
57 <60> send: +
58 <D0> popIntoTemp: 0
59 <41> pushTemp: 1
60 <51> pushConstant: 1
61 <60> send: +
62 <D1> popIntoTemp: 1
63 <E1 FF ED ED> jumpTo: 48
67 <40> pushTemp: 0
68 <5C> returnTop
```

```
Object>>at: index
<primitive: 60>
^self error
```

```
<primitive: 60>
33 <F8 3C 00> callPrimitive: 60
36 <4C> self
37 <80> send: error
38 <5C> returnTop
```

# Background: OSVM – JIT



```
000d2698: xorq %rdx, %rdx : 48 31 D2
000d269b: call .-0xD1DF8 (0x8a8=ceMethodAbort0Args) : E8 08 E2 F2 FF
entry:
000d26a0: movq %rdx, %rax : 48 89 D0
000d26a3: andq $0x7, %rax : 48 83 E0 07
000d26a7: jnz .+0x9 (.+0042) : 75 09
000d26a9: movq (%rdx), %rax : 48 8B 02
000d26ac: andq $0x3ffff, %rax : 48 25 FF FF 3F 00
000d26b2: cmpq %rcx, %rax : 48 39 C8
000d26b5: jnz .-0x1C (-+002B) : 75 E4
noCheckEntry:
000d26b7: movq (%rsp), %r9 : 4C 8B 0C 24
000d26bb: movq %rdx, (%rsp) : 48 89 14 24
000d26bf: pushq %r9 : 41 51
000d26c1: pushq %rbp : 55
000d26c2: movq %rsp, %rbp : 48 89 E5
000d26c5: leaq 0xffffffffffa4(%rip), %r8 = sum3@0 : 4C 8D 05 A4 FF FF FF
000d26cc: pushq %r8 : 41 50
IsAbsPCReference:
000d26ce: movq $0x8aee08=nil, %r9 : 49 C7 C1 08 EE 8A 00
000d26d5: pushq %r9 : 41 51
000d26d7: pushq %rdx : 52
000d26d8: pushq %r9 : 41 51
000d26da: pushq %r9 : 41 51
000d26dc: pushq %r9 : 41 51
000d26de: movq (%rbx), %rax = &stackLimitFromMachineCode : 48 8B 03
000d26e1: cmpq %rax, %rsp : 48 39 C4
000d26e4: jb .-0x4E (-+0028) : 72 B2
HasBytecodePC/StackCheck:
000d26e6: movq $0x1, %rax : 48 C7 C0 01 00 00 00
000d26ed: movq %rax, -32(%rbp) : 48 89 45 E0
000d26f1: movq $0x9, %rax : 48 C7 C0 09 00 00 00
000d26f8: movq %rax, -40(%rbp) : 48 89 45 D8
000d26fc: movq -24(%rbp), %rdx : 48 8B 55 E8
000d2700: movq 0x33=Array, %rcx : 48 C7 C1 33 00 00 00
000d2707: call .-0xCD85C (0x4eb0=size@30) : E8 A4 27 F3 FF
IsSendCall size bc 46/47:
000d270c: movq %rdx, -48(%rbp) : 48 89 55 D0
000d2710: movq -48(%rbp), %rdi : 48 8B 7D D0
000d2714: movq -40(%rbp), %rdx : 48 8B 55 D8
000d2718: movq %rdx, %rax : 48 89 D0
000d271b: andq %rdi, %rax : 48 23 C7
000d271e: testb $0x01, %al : 48 F6 C0 01
000d2722: jz .+0xB (-+00BF) : 74 0B
000d2724: cmpq %rdi, %rdx : 48 39 FA
000d2727: jnle .+0x96 (-+0153) : 0F 8F 96 00 00 00
000d272d: jmp .+0x26 (-+00E5) : EB 26
000d272f: movq 0xfffffffffffb=#<=, %rcx : 48 C7 C1 FB FF FF FF
000d2736: call .-0xD227B (0x4c0=ceSend1Args) : E8 85 DD F2 FF
IsSendCall <= bc 50/51:
000d273b: movq %rdx, %rax : 48 89 D0
000d273e: subq $0x8aee18=false, %rax : 48 2D 18 EE 8A 00
000d2744: jz .+0x79 (-+0153) : 0F 84 79 00 00 00
000d274a: cmpq $0x10, %rax : 48 83 F8 10
000d274e: jz .+0x5 (-+00E5) : 74 05
000d2750: call .-0xD11F5 (0x1560=ceSendMustBeBooleanAddFalseTrampoline) : E8 0B EE F2 FF
IsRelativeCall:
HasBytecodePC bc 52/53:
000d2755: movq -32(%rbp), %rax : 48 8B 45 E0
```

```
000d2759: pushq %rax : 50
000d275a: movq -40(%rbp), %rdi : 48 8B 7D D8
000d275e: movq -24(%rbp), %rdx : 48 8B 55 E8
000d2762: movq 0x33=Array, %rcx : 48 C7 C1 33 00 00 00
000d2769: call .-0x4FDE (0xcd790=at2:@30) : E8 22 B0 FF FF
IsSendCall at: bc 56/57:
000d276e: movq %rdx, %rdi : 48 89 D7
000d2771: popq %rdx : 5A
000d2772: movq 0x1=SmallInteger, %rcx : 48 C7 C1 01 00 00 00
000d2779: call .-0xCEC7E (0x3b00=+@30) : E8 82 13 F3 FF
IsSendCall + bc 57/58:
000d277e: movq %rdx, -32(%rbp) : 48 89 55 E0
000d2782: movq -40(%rbp), %rdx : 48 8B 55 D8
000d2786: testb $0x01, %dil : 48 F6 C2 01
000d278a: jz .+0xA (-+0126) : 74 0A
000d278c: addq $0x8, %rdx : 48 83 C2 08
000d2790: jno .+0x17 (-+0139) : 71 17
000d2792: subq $0x8, %rdx : 48 83 EA 08
000d2796: movq $0x9, %rdi : 48 C7 C7 09 00 00 00
000d279d: movq 0xffffffffffff=#+, %rcx : 48 C7 C1 FF FF FF FF
000d27a4: call .-0xD22E9 (0x4c0=ceSend1Args) : E8 17 DD F2 FF
IsSendCall + bc 61/62:
000d27a9: movq %rdx, -40(%rbp) : 48 89 55 D8
000d27ad: movq (%rbx), %rax = &stackLimitFromMachineCode : 48 8B 03
000d27b0: cmpq %rax, %rsp : 48 39 C4
000d27b3: jnb .-0xA9 (-+00A0) : 0F 83 57 FF FF FF
000d27b9: call .-0xD11CE (0x15f0=ceCheckForInterruptTrampoline) : E8 32 EE F2 FF
IsRelativeCall:
HasBytecodePC bc 62/63:
000d27be: jmp .-0xB3 (-+00A0) : E9 4D FF FF FF
000d27c3: movq -32(%rbp), %rdx : 48 8B 55 E0
000d27c7: movq %rbp, %rsp : 48 89 EC
000d27ca: popq %rbp : 5D
000d27cb: ret $0x0008 : C2 08 00
000d27ce: int3 : CC
000d27cf: int3 : CC
000d27d0: int3 : CC
```

```
000cd788: xorq %rdx, %rdx : 48 31 D2
000cd78b: call .-0xCCE78 (0x918=ceMethodAbort1Args) : E8 88 31 F3 FF
entry:
000cd790: movq %rdx, %rax : 48 89 D0
000cd793: andq $0x7, %rax : 48 83 E0 07
000cd797: jnz .+0x9 (-+0042) : 75 09
000cd799: movq (%rdx), %rax : 48 8B 02
000cd79c: andq $0x3ffff, %rax : 48 25 FF FF 3F 00
000cd7a2: cmpq %rcx, %rax : 48 39 C8
000cd7a5: jnz .-0x1C (-+002B) : 75 E4
noCheckEntry:
000cd7a7: testb $0x07, %dil : 48 F6 C2 07
000cd7ab: jnz .+0x1CA (-+021B) : 0F 85 CA 01 00 00
000cd7b1: movq %rdi, %rsi : 48 89 FE
000cd7b4: testb $0x01, %dil : 48 F6 C7 01
000cd7b8: jz .+0x1BD (-+021B) : 0F 84 BD 01 00 00
000cd7be: sarq $0x03, %rsi : 48 C1 FE 03
000cd7c2: subq $0x1, %rsi : 48 83 EE 01
000cd7c6: movq (%rdx), %rax : 48 8B 02
000cd7c9: movq %rax, %r9 : 49 89 C1
000cd7cc: shrq $0x18, %r9 : 49 C1 E9 18
000cd7d0: andq $0x1f, %r9 : 49 83 E1 1F
000cd7d4: xorq %rcx, %rcx : 48 31 C9
000cd7d7: movb 0x7(%rdx), %cl : 48 8A 4A 07
000cd7db: cmpq $0xff, %rcx : 48 81 F9 FF 00 00 00
000cd7e2: jl .+0xC (-+0090) : 7C 0C
000cd7e4: movq 0xffffffffffff(%rdx), %rcx : 48 8B 4A F8
000cd7e8: shlq $0x08, %rcx : 48 C1 E1 08
000cd7ec: shrq $0x08, %rcx : 48 C1 E9 08
000cd7f0: cmpq $0x10, %r9 : 49 83 F9 10
000cd7f4: jnb .+0x3F (-+00D5) : 73 3F
000cd7f6: cmpq $0x2, %r9 : 49 83 F9 02
000cd7fa: jz .+0x27 (-+00C3) : 74 27
000cd7fc: jb .+0x20 (-+00BE) : 72 20
000cd7fe: cmpq $0x4, %r9 : 49 83 F9 04
000cd802: jbe .+0xA5 (-+014D) : 0F 86 A5 00 00 00
000cd808: cmpq $0xc, %r9 : 49 83 F9 0C
000cd80c: jnb .+0x5C (-+010A) : 73 5C
000cd80e: cmpq $0xa, %r9 : 49 83 F9 0A
000cd812: jnb .+0x77 (-+012B) : 73 77
000cd814: cmpq $0x9, %r9 : 49 83 F9 09
000cd818: jz .+0xEF (-+01AD) : 0F 84 EF 00 00 00
000cd81e: jmp .+0x158 (-+021B) : E9 58 01 00 00
000cd823: cmpq %rsi, %rcx : 48 39 F1
000cd826: jbe .+0x14F (-+021B) : 0F 86 4F 01 00 00
000cd82c: addq $0x1, %rsi : 48 83 C6 01
000cd830: movq (%rdx,%rsi,8), %rdx : 48 8B 14 F2
000cd834: ret : C3
000cd835: shlq $0x03, %rcx : 48 C1 E1 03
000cd839: movq %r9, %rax : 4C 89 C8
000cd83c: andq $0x7, %rax : 48 83 E0 07
000cd840: subq %rax, %rcx : 48 2B C8
000cd843: cmpq %rsi, %rcx : 48 39 F1
000cd846: jbe .+0x12F (-+021B) : 0F 86 2F 01 00 00
000cd84c: cmpq $0x18, %r9 : 49 83 F9 18
000cd850: jnb .+0x103 (-+01F9) : 0F 83 03 01 00 00
000cd856: addq $0x8, %rsi : 48 83 C6 08
000cd85a: movb (%rdx,%rsi,1), %r9b : 4C 8A 0C 32
000cd85e: movq %r9, %rdx : 4C 89 CA
000cd861: shlq $0x03, %rdx : 48 C1 E2 03
000cd865: addq $0x1, %rdx : 48 83 C2 01
000cd869: ret : C3
000cd86a: shlq $0x02, %rcx : 48 C1 E1 02
```



# Background: OSVM – Simulator



Simulation of spurreader-64.image

Projects Tools Apps Do Extras Windows Help HomeProject Search or evaluate...

**Transcript**

-----QUIT-----{24 April 2025 . 1:39:32 pm} spurreader-64.image priorSource: 3859657

Input Requested

Input please!

Accept Cancel

Reading a ReadStream

436,000/16,500

selctr: A89E70 #center  
blkentry: 0  
stackCheckOffset: 70/863F0

000863a8: xorq %rdx, %rdx : 48 31 D2  
000863ab: call .-0x85B08 (0x8a8=ceMethodAbort0Args) : E8 F8 A4 F7 FF  
entry:  
000863b0: movq %rdx, %rax : 48 89 D0  
000863b3: andq \$0x7, %rax : 48 83 E0 07  
000863b7: jnz .+0x9 (.+0042) : 75 09  
000863b9: movq (%rdx), %rax : 48 8B 02  
000863bc: andq \$0x3ffff, %rax : 48 25 FF FF 3F 00  
000863c2: cmpq %rcx, %rax : 48 39 C8  
000863c5: jnz .-0x1C (.+002B) : 75 E4  
noCheckEntry:  
000863c7: movq (%rsp), %r9 : 4C 8B 0C 24  
000863cb: movq %rdx, (%rsp) : 48 89 14 24  
000863cf: pushq %r9 : 41 51

toggle transcript  
deliver interrupt  
clone VM  
print head frame  
short print head frame & callers  
long print head frame & callers  
print frame...  
inspect frame...  
print call stack  
print stack call stack  
print stack call stack of...  
print call stack of...  
print call stack of frame...  
print all stacks  
write back local ptrs  
write back mc ptrs  
print rump C stack  
print registers  
disassemble method/trampoline...  
disassemble method/trampoline at pc  
disassemble ext head frame method  
print oop...  
long print oop...  
print context...  
inspect object memory  
run leak checker  
inspect cointerpreter  
inspect cogit  
inspect method zone  
inspect head frame  
inspect processor  
print cog methods  
print cog methods with prim...  
print cog methods with selector...  
print cog methods with method...  
print cog method for...  
print cog method header for...  
print trampoline table  
print prim trace log  
report recent instructions  
print registers each instruction  
print instructions each instruction  
print bytecode each bytecode  
print frame each bytecode  
single step  
click step  
break on execute method  
set break pc...-ve to disable or remove  
set break count...  
set break selector...  
set break block in cointerpreter...  
set break block in cogit...  
set cogit break method...

# Background: OSVM – First Contributions



System Browser: StackInterpreter

Building

Interpreter

InterpreterSimulation

JIT

JITSimulation

StackInterpreter

StackInterpreterPrimitive

VMBasicConstants

-- all --

-- core --

-- extensions --

I/O primitive support

I/O primitives

interpret

interpreterAllocationRese

ioBeDisplay:width:height:

ioCanCatchFFIExceptions

ioFilename:fromString:ofl

instance

class

?

browse

senders

implementors

versions

inheritance

hierarchy

variables

source

interpret

"This is the main interpreter loop. It normally loops forever, fetching and executing bytecodes. When running in the context of a browser plugin VM, however, it must return control to the browser periodically. This should done only when the state of the moment that a checks happen"

<inline: false>

"If stacklimit"

stackLimit =

[^self int

"record entry

self browser

self setMeth

self assertVa

self internali

self initExter

self fetchNex

[true] whileT

[self disp

localIP := localIP - 1.

undo the pre-increment

self externalizePandSP.

^nil

Changes go to "fix-primitiveDoPrimitiveWithArgs"

fix-primitiveDoPrimitiveWithArgs

GStreamer-Plugins-JMM.16

primitiveDoPrimitiveWithArgs {control primitives}

primitiveFunctionPointer := self functionPointerFor: primIdx inClass: nil.

primitiveFunctionPointer = 0 ifTrue:

[primitiveFunctionPointer := #primitiveDoPrimitiveWithArgs.

^self primitiveFailFor: PrimErrBadIndex negated].

Revise installation instructions for Ubuntu #716

Merged

eliotmiranda merged 1 commit into OpenSmalltalk:Cog from LinqLover:docs/ubuntu-HowToBuild-details 3 weeks ago

Conversation 0

LinqLover

Add support for clang>=16 [v2] #719

Merged

eliotmiranda merged 5 commits into OpenSmalltalk:Cog from MariusDoe:clang16-v2 2 weeks ago

Conversation 3

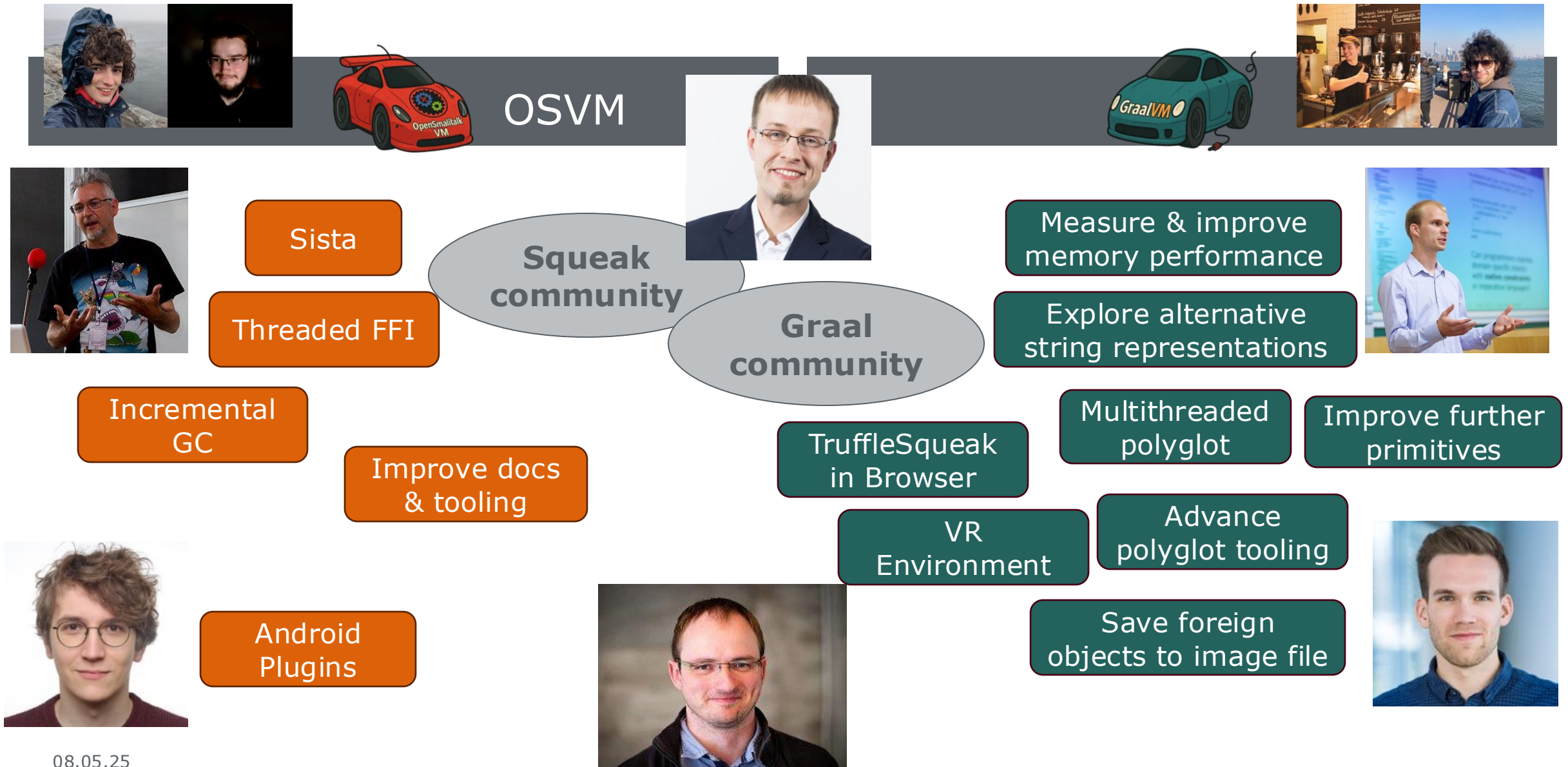
Commits 5

Checks 0

Files changed 14

MariusDoe commented 2 weeks ago • edited

# Ideas: Stakeholders and Collection



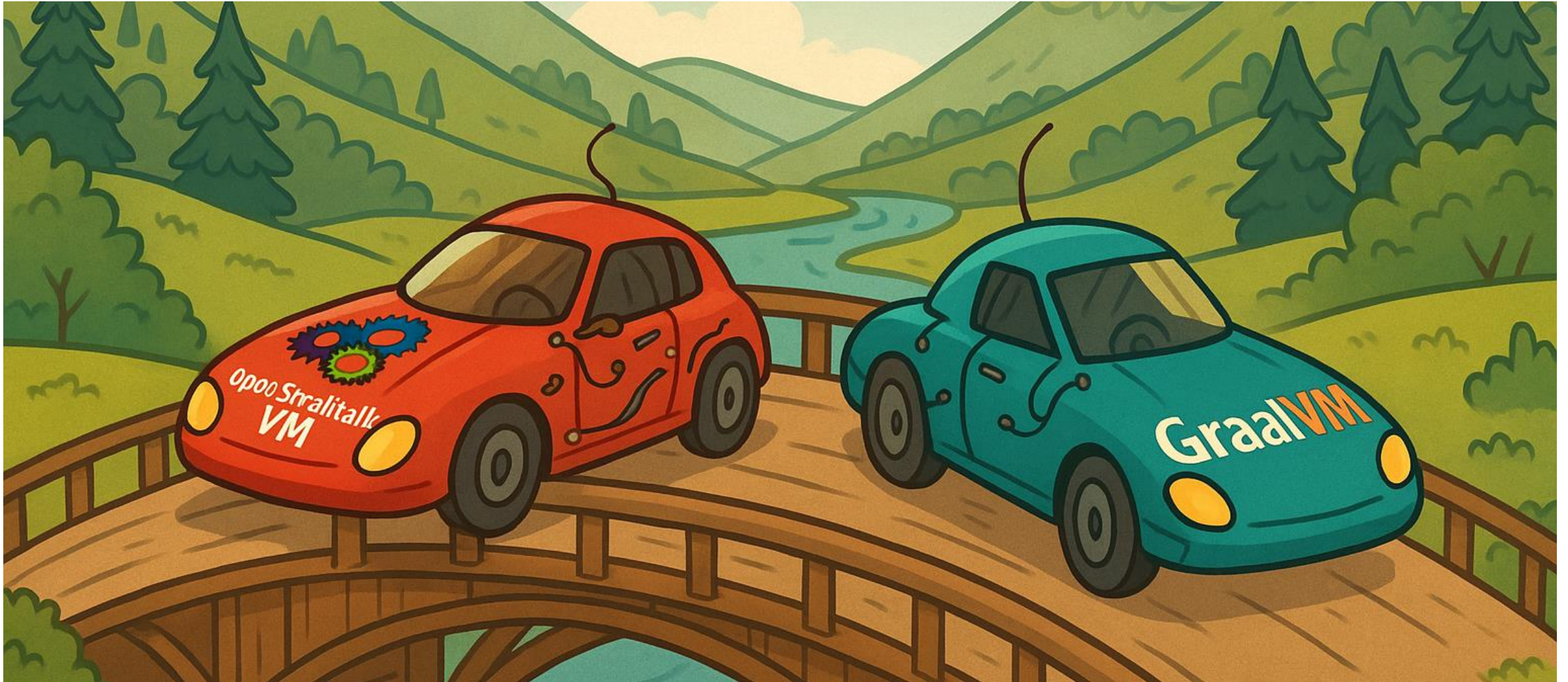
# Ideas: Possible Research Questions



- How do both VMs differ in **supporting** us while **building tools** that **cross system boundaries**?
- *Which VM offers more suitable **abstractions, workflows, and tooling** for **exploring, developing, and measuring** them, and what can they learn from each other?*
- *Which VM offers better **runtime performance** under typical Smalltalk workloads?*
- *How do **memory usage patterns** differ between both VMs under typical Smalltalk workloads?*



# Next Steps



# Next Steps: OSVM – Sista



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| <pre>000d2698: xorq %rdx, %rdx : 48 31 D2 000d269b: call -.0xD1DF8 (0x8a8=ceMethodAbort0Args) : E8 08 E2 F2 FF <b>entry:</b> 000d26a0: movq %rdx, %rax : 48 89 D0 000d26a3: andq \$0x7, %rax : 48 83 E0 07 000d26a7: jnz .+0x9 (.+0042) : 75 09 000d26a9: movq (%rdx), %rax : 48 8B 02 000d26ac: andq \$0x3fffff, %rax : 48 25 FF FF 3F 00 000d26b2: cmpq %rcx, %rax : 48 39 C8 000d26b5: jnz -.0x1C (-.002B) : 75 E4 <b>noCheckEntry:</b> 000d26b7: movq (%rsp), %r9 : 4C 8B 0C 24 000d26bb: movq %rdx, (%rsp) : 48 89 14 24 000d26bf: pushq %r9 : 41 51 000d26c1: pushq %rbp : 55 000d26c2: movq %rsp, %rbp : 48 89 E5 000d26c5: leaq 0xffffffffffa4(%rip), %r8 = sum3@0 : 4C 8D 05 A4 FF FF FF 000d26cc: pushq %r8 : 41 50 <b>IsAbsPCReference:</b> 000d26ce: movq \$0x8aee08=nil, %r9 : 49 C7 C1 08 EE 8A 00 000d26d5: pushq %r9 : 41 51 000d26d7: pushq %rdx : 52 000d26d8: pushq %r9 : 41 51 000d26da: pushq %r9 : 41 51 000d26dc: pushq %r9 : 41 51 000d26de: movq (%rbx), %rax = &amp;stackLimitFromMachineCode : 48 8B 03 000d26e1: cmpq %rax, %rsp : 48 39 C4 000d26e4: jb -.0x4E (.+0028) : 72 B2 <b>HasBytecodePC/StackCheck:</b> 000d26e6: movq \$0x1, %rax : 48 C7 C0 01 00 00 00 000d26ed: movq %rax, -32(%rbp) : 48 89 45 E0 000d26f1: movq \$0x9, %rax : 48 C7 C0 09 00 00 00 000d26f8: movq %rax, -40(%rbp) : 48 89 45 D8 000d26fc: movq -24(%rbp), %rdx : 48 8B 55 E8 000d2700: movq 0x33=Array, %rcx : 48 C7 C1 33 00 00 00 000d2707: call -.0xCD85C (0x4eb0=size@30) : E8 A4 27 F3 FF <b>IsSendCall size bc 46/47:</b> 000d270c: movq %rdx, -48(%rbp) : 48 89 55 D0 000d2710: movq -48(%rbp), %rdi : 48 8B 7D D0 000d2714: movq -40(%rbp), %rdx : 48 8B 55 D8 000d2718: movq %rdx, %rax : 48 89 D0 000d271b: andq %rdi, %rax : 48 23 C7 000d271e: testb \$0x01, %al : 48 F6 C0 01 000d2722: jz .+0xB (.+00BF) : 74 0B 000d2724: cmpq %rdi, %rdx : 48 39 FA 000d2727: jnle .+0x96 (.+0153) : 0F 8F 96 00 00 00 000d272d: jmp .+0x26 (.+00E5) : EB 26 000d272f: movq 0xfffffffffffb=#&lt;=, %rcx : 48 C7 C1 FB FF FF FF 000d2736: call -.0xD227B (0x4c0=ceSend1Args) : E8 85 DD F2 FF <b>IsSendCall &lt;= bc 50/51:</b> 000d273b: movq %rdx, %rax : 48 89 D0 000d273e: subq \$0x8aee18=false, %rax : 48 2D 18 EE 8A 00 000d2744: jz .+0x79 (.+0153) : 0F 84 79 00 00 00 000d274a: cmpq \$0x10, %rax : 48 83 F8 10 000d274e: jz .+0x5 (.+00E5) : 74 05 000d2750: call -.0xD11F5 (0x1560=ceSendMustBeBooleanAddFalseTrampoline) : E8 0B EE F2 FF <b>IsRelativeCall:</b> <b>HasBytecodePC bc 52/53:</b> 000d2755: movq -32(%rbp), %rax : 48 8B 45 E0</pre> | <pre>000d2759: pushq %rax : 50 000d275a: movq -40(%rbp), %rdi : 48 8B 7D D8 000d275e: movq -24(%rbp), %rdx : 48 8B 55 E8 000d2762: movq 0x33=Array, %rcx : 48 C7 C1 33 00 00 00 000d2769: call -.0x4FDE (0xcd790=at2:@30) : E8 22 B0 FF FF <b>IsSendCall at: bc 56/57:</b> 000d276e: movq %rdx, %rdi : 48 89 D7 000d2771: popq %rdx : 5A 000d2772: movq 0x1=SmallInteger, %rcx : 48 C7 C1 01 00 00 00 000d2779: call -.0xCEC7E (0x3b00=+@30) : E8 82 13 F3 FF <b>IsSendCall + bc 57/58:</b> 000d277e: movq %rdx, -32(%rbp) : 48 89 55 E0 000d2782: movq -40(%rbp), %rdx : 48 8B 55 D8 000d2786: testb \$0x01, %dil : 48 F6 C2 01 000d278a: jz .+0xA (.+0126) : 74 0A 000d278c: addq \$0x8, %rdx : 48 83 C2 08 000d2790: jno .+0x17 (.+0139) : 71 17 000d2792: subq \$0x8, %rdx : 48 83 EA 08 000d2796: movq \$0x9, %rdi : 48 C7 C7 09 00 00 00 000d279d: movq 0xffffffffffff=#+, %rcx : 48 C7 C1 FF FF FF FF 000d27a4: call -.0xD22E9 (0x4c0=ceSend1Args) : E8 17 DD F2 FF <b>IsSendCall + bc 61/62:</b> 000d27a9: movq %rdx, -40(%rbp) : 48 89 55 D8 000d27ad: movq (%rbx), %rax = &amp;stackLimitFromMachineCode : 48 8B 03 000d27b0: cmpq %rax, %rsp : 48 39 C4 000d27b3: jnb -.0xA9 (.+00A0) : 0F 83 57 FF FF FF 000d27b9: call -.0xD11CE (0x15f0=ceCheckForInterruptTrampoline) : E8 32 EE F2 FF <b>IsRelativeCall:</b> <b>HasBytecodePC bc 62/63:</b> 000d27be: jmp -.0xB3 (.+00A0) : E9 4D FF FF FF 000d27c3: movq -32(%rbp), %rdx : 48 8B 55 E0 000d27c7: movq %rbp, %rsp : 48 89 EC 000d27ca: popq %rbp : 5D 000d27cb: ret \$0x0008 : C2 08 00 000d27ce: int3 : CC 000d27cf: int3 : CC 000d27d0: int3 : CC</pre> |
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| <pre>000cd788: xorq %rdx, %rdx : 48 31 D2 000cd78b: call -.0xCCE78 (0x918=ceMethodAbort1Args) : E8 88 31 F3 FF <b>entry:</b> 000cd790: movq %rdx, %rax : 48 89 D0 000cd793: andq \$0x7, %rax : 48 83 E0 07 000cd797: jnz .+0x9 (.+0042) : 75 09 000cd799: movq (%rdx), %rax : 48 8B 02 000cd79c: andq \$0x3fffff, %rax : 48 25 FF FF 3F 00 000cd7a2: cmpq %rcx, %rax : 48 39 C8 000cd7a5: jnz -.0x1C (-.002B) : 75 E4 <b>noCheckEntry:</b> 000cd7a7: testb \$0x07, %dil : 48 F6 C2 07 000cd7ab: jnz .+0x1CA (.+021B) : 0F 85 CA 01 00 00 000cd7b1: movq %rdi, %rsi : 48 89 FE 000cd7b4: testb \$0x01, %dil : 48 F6 C7 01 000cd7b8: jz .+0x1BD (.+021B) : 0F 84 BD 01 00 00 000cd7be: sarq \$0x03, %rsi : 48 C1 FE 03 000cd7c2: subq \$0x1, %rsi : 48 83 EE 01 000cd7c6: movq (%rdx), %rax : 48 8B 02 000cd7c9: movq %rax, %r9 : 49 89 C1 000cd7cc: shrq \$0x18, %r9 : 49 C1 E9 18 000cd7d0: andq \$0x1f, %r9 : 49 83 E1 1F 000cd7d4: xorq %rcx, %rcx : 48 31 C9 000cd7d7: movb 0x7(%rdx), %cl : 48 8A 4A 07 000cd7db: cmpq \$0xff, %rcx : 48 81 F9 FF 00 00 00 000cd7e2: jl .+0xC (.+0090) : 7C 0C 000cd7e4: movq 0xffffffffffff(%rdx), %rcx : 48 8B 4A F8 000cd7e8: shlq \$0x08, %rcx : 48 C1 E1 08 000cd7ec: shrq \$0x08, %rcx : 48 C1 E9 08 000cd7f0: cmpq \$0x10, %r9 : 49 83 F9 10 000cd7f4: jnb .+0x3F (.+00D5) : 73 3F 000cd7f6: cmpq \$0x2, %r9 : 49 83 F9 02 000cd7fa: jz .+0x27 (.+00C3) : 74 27 000cd7fc: jb .+0x20 (.+00BE) : 72 20 000cd7fe: cmpq \$0x4, %r9 : 49 83 F9 04 000cd802: jbe .+0xA5 (.+014D) : 0F 86 A5 00 00 00 000cd808: cmpq \$0xc, %r9 : 49 83 F9 0C 000cd80c: jnb .+0x5C (.+010A) : 73 5C 000cd80e: cmpq \$0xa, %r9 : 49 83 F9 0A 000cd812: jnb .+0x77 (.+012B) : 73 77 000cd814: cmpq \$0x9, %r9 : 49 83 F9 09 000cd818: jz .+0xEf (.+01AD) : 0F 84 EF 00 00 00 000cd81e: jmp .+0x158 (.+021B) : E9 58 01 00 00 000cd823: cmpq %rsi, %rcx : 48 39 F1 000cd826: jbe .+0x14F (.+021B) : 0F 86 4F 01 00 00 000cd82c: addq \$0x1, %rsi : 48 83 C6 01 000cd830: movq (%rdx,%rsi,8), %rdx : 48 8B 14 F2 000cd834: ret : C3 <b>000cd835: shlq \$0x03, %rcx : 48 C1 E1 03</b> 000cd839: movq %r9, %rax : 4C 89 C8 000cd83c: andq \$0x7, %rax : 48 83 E0 07 000cd840: subq %rax, %rcx : 48 2B C8 000cd843: cmpq %rsi, %rcx : 48 39 F1 000cd846: jbe .+0x12F (.+021B) : 0F 86 2F 01 00 00 000cd84c: cmpq \$0x18, %r9 : 49 83 F9 18 000cd850: jnb .+0x103 (.+01F9) : 0F 83 03 01 00 00 000cd856: addq \$0x8, %rsi : 48 83 C6 08 000cd85a: movb (%rdx,%rsi,1), %r9b : 4C 8A 0C 32 000cd85e: movq %r9, %rdx : 4C 89 CA 000cd861: shlq \$0x03, %rdx : 48 C1 E2 03 000cd865: addq \$0x1, %rdx : 48 83 C2 01 000cd869: ret : C3 <b>000cd86a: shlq \$0x02, %rcx : 48 C1 E1 02</b></pre> |
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```

D D8
55 E8
C1 33 00 00 00
0) : E8 22 B0 FF FF

48 C7 C1 01 00 00 00
: E8 82 13 F3 FF

55 E0
55 D8

00 00 00
48 C7 C1 FF FF FF FF
Args) : E8 17 DD F2 FF

55 D8
hitFromMachineCode : 48 8B 03

FF FF FF
(ForInterruptTrampoline) : E8 32 EE F2 FF

FF FF
55 E0

```

```

000cd788: xorq %rdx, %rdx : 48 31 D2
000cd78b: call -.0xCCE78 (0x918=ceMethodAbort1Args) : E8 88 31 F3 FF
entry:
000cd790: movq %rdx, %rax : 48 89 D0
000cd793: andq $0x7, %rax : 48 83 E0 07
000cd797: jnz .+0x9 (.+0042) : 75 09
000cd799: movq (%rdx), %rax : 48 8B 02
000cd79c: andq $0x3ffff, %rax : 48 25 FF FF 3F 00
000cd7a2: cmpq %rcx, %rax : 48 39 C8
000cd7a5: jnz -.0x1C (.+002B) : 75 E4
noCheckEntry:
000cd7a7: testb $0x07, %dil : 48 F6 C2 07
000cd7ab: jnz .+0x1CA (.+021B) : 0F 85 CA 01 00 00
000cd7b1: movq %rdi, %rsi : 48 89 FE
000cd7b4: testb $0x01, %dil : 48 F6 C7 01
000cd7b8: jz .+0x1BD (.+021B) : 0F 84 BD 01 00 00
000cd7be: sarq $0x03, %rsi : 48 C1 FE 03
000cd7c2: subq $0x1, %rsi : 48 83 EE 01
000cd7c6: movq (%rdx), %rax : 48 8B 02
000cd7c9: cmpq %rax, %r9 : 49 89 C1
000cd7cc: shrq $0x18, %r9 : 49 C1 E9 18
000cd7d0: andq $0x1f, %r9 : 49 83 E1 1F
000cd7d4: xorq %rcx, %rcx : 48 31 C9
000cd7d7: movb 0x7(%rdx), %dil : 48 8A 4A 07
000cd7db: cmpq $0xff, %rcx : 48 81 F9 FF 00 00 00
000cd7e2: jl -.0xC (.+0090) : 7C 0C
000cd7e4: movq 0xffffffffffff8(%rdx), %rcx : 48 8B 4A F8
000cd7e8: shlq $0x08, %rcx : 48 C1 E1 08
000cd7ec: shrq $0x08, %rcx : 48 C1 E9 08
000cd7f0: cmpq $0x10, %r9 : 49 83 F9 10
000cd7f4: jnb .+0x3F (.+00D5) : 73 3F
000cd7f6: cmpq $0x2, %r9 : 49 83 F9 02
000cd7fa: jz .+0x27 (.+00C3) : 74 27
000cd7fc: jb .+0x20 (.+00BE) : 72 20
000cd7fe: cmpq $0x4, %r9 : 49 83 F9 04
000cd802: jbe .+0xA5 (.+014D) : 0F 86 A5 00 00 00
000cd808: cmpq $0xc, %r9 : 49 83 F9 0C
000cd80c: jnb .+0x5C (.+010A) : 73 5C
000cd80e: cmpq $0xa, %r9 : 49 83 F9 0A
000cd812: jnb .+0x77 (.+012B) : 73 77
000cd814: cmpq $0x9, %r9 : 49 83 F9 09
000cd818: jz .+0xEf (.+01AD) : 0F 84 EF 00 00 00
000cd81e: jmp .+0x158 (.+021B) : E9 58 01 00 00
000cd823: cmpq %rsi, %rcx : 48 39 F1
000cd826: jbe .+0x14F (.+021B) : 0F 86 4F 01 00 00
000cd82c: addq $0x1, %rsi : 48 83 C6 01
000cd830: movq (%rdx,%rsi,8), %rdx : 48 8B 14 F2
000cd834: ret : C3
000cd835: shlq $0x03, %rcx : 48 C1 E1 03
000cd839: movq %r9, %rax : 48 C9 C8
000cd83c: andq $0x7, %rax : 48 83 E0 07
000cd840: subq %rax, %rcx : 48 2B C8
000cd843: cmpq %rsi, %rcx : 48 39 F1
000cd846: jbe .+0x12F (.+021B) : 0F 86 2F 01 00 00
000cd84c: cmpq $0x18, %r9 : 49 83 F9 18
000cd850: jnb .+0x103 (.+01F9) : 0F 83 03 01 00 00
000cd856: addq $0x8, %rsi : 48 83 C6 08
000cd85a: movb (%rdx,%rsi,1), %r9b : 4C 8A 0C 32
000cd85e: movq %r9, %rdx : 4C 89 CA
000cd861: shlq $0x03, %rdx : 48 C1 E2 03
000cd865: addq $0x1, %rdx : 48 83 C2 01
000cd869: ret : C3
000cd86a: shlq $0x02, %rcx : 48 C1 E1 02

```

# Next Steps: OSVM – Sista



## Context:

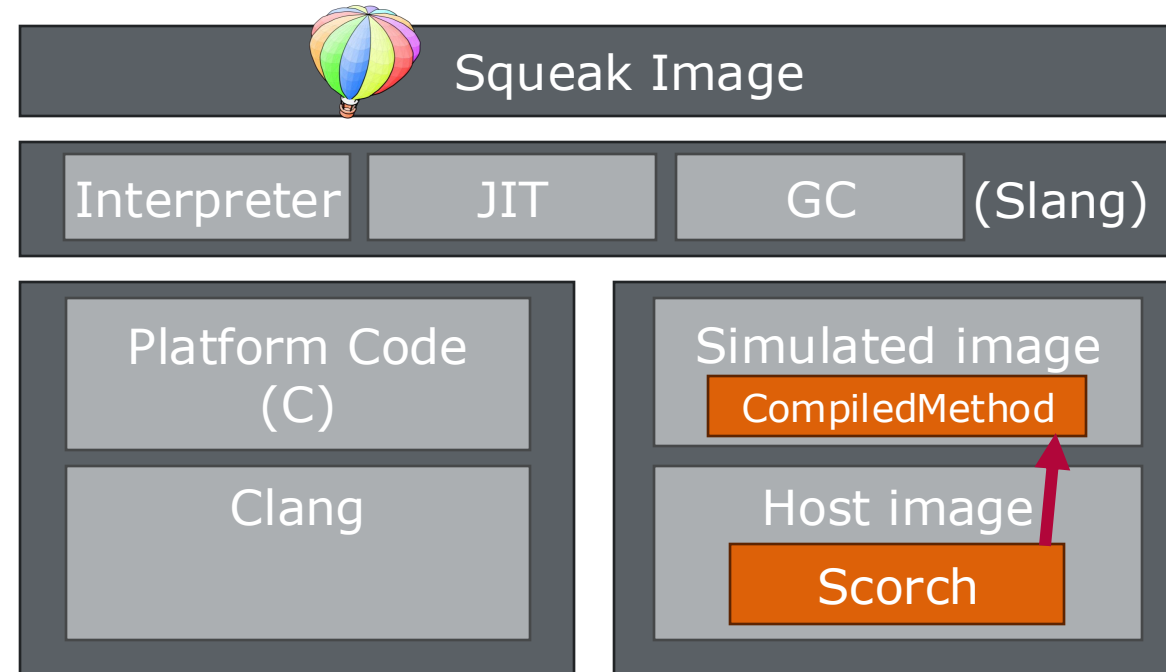
- Many **redundant checks** in jitted assembly code
- Approach:** Sista (Speculative Inlining SmallTalk Architecture)
- VM provides **statistics** about frequently executed code branches
- **Adaptive optimizer** in image (*Scorch*) optimizes critical methods by **inlining** relevant methods
- Inlined methods contain **unsafe bytecodes** that are jitted without checks

## Problem:

- Initial bugs in adaptive optimizer **crash** the VM -> **practical development** of optimizer not possible

## Intended solution:

- Test and debug adaptive optimizations in **simulator**
- Run adaptive optimizer in **host VM**
- Bridge **simulation boundary** to operate on methods and stats of simulated image





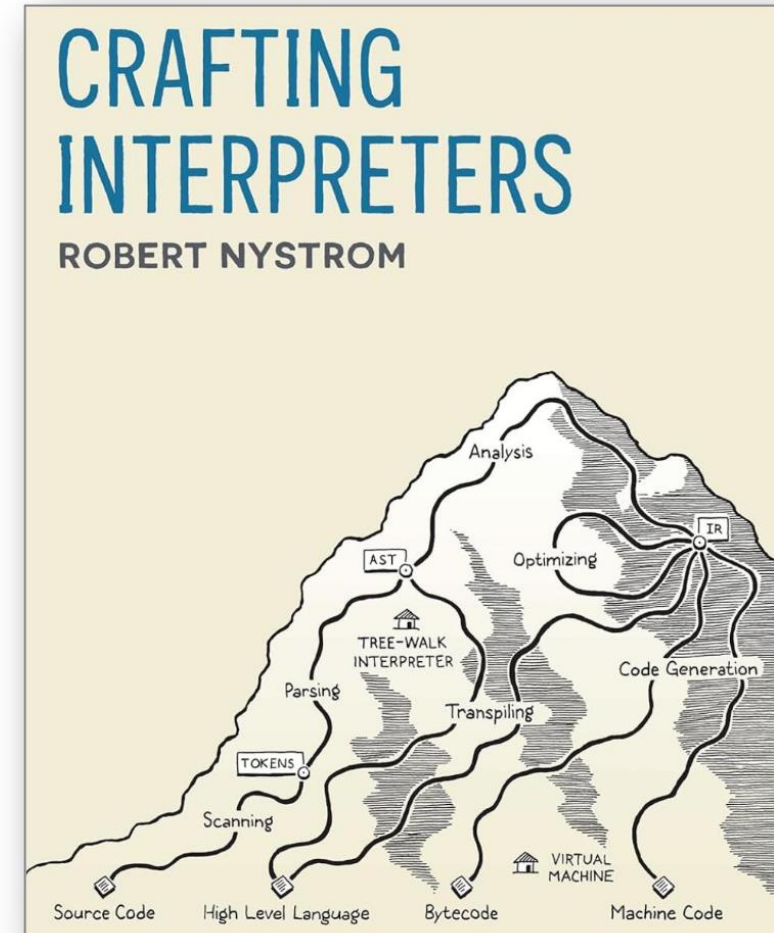
# Next Steps: TSVM



## Work with polyglot tooling in TruffleSqueak

1. Integrate lox language from BYOPL seminar into TruffleSqueak
2. Gather insights about tooling and write them down in developer journal
3. Extract ideas from insights, literature and interviews

How can we detect with **polyglot tooling** in TruffleSqueak **inefficient implementations** in truffle interpreters?



[Nystrom2021]

Robert Nystrom. *Crafting Interpreters*, 2021

<https://craftinginterpreters.com/>

# Discussion: Our (and your) next questions

1. How and for which user group can we compare tooling and developer experience?
2. How can we integrate more polyglot programming tools in TSVM?
3. How could we design rich tooling for understanding profiling data for developing the Sista optimizer?
4. Which other aspects can be worth exploring for comparing both VMs?



# Stakeholders (Hidden)



| Stakeholder     | VM   | Interested in?                                                      | Object of Communication                                         |
|-----------------|------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| Marcel Taeumel  | Both | Comparing OSVM vs TSVM; internal collaboration and learning process | Project mentoring and prioritization                            |
| Jens Lincke     | TSVM | GraalVM & Truffle, Memory consumption, compare VR performance       | Project mentoring                                               |
| Fabio Niephaus  | TSVM | Improving TruffleSqueak in any way, community                       | Provide knowledge about TruffleSqueak                           |
| Tim Felgentreff | TSVM | GraalVM & Truffle, Memory consumption                               | Provide knowledge about GraalVM                                 |
| Eliot Miranda   | OSVM | Making OSVM the fastest Smalltalk VM until 2028 (GC, FFI, JIT)      | Provide knowledge about OSVM                                    |
| (Leonard Geier) | OSVM | Using OSVM for VR (performance, minimize lags)                      | Could provide practical benchmarks for performance improvements |